

Closing the Gender Gap?

The Employment Effects of 3G Internet in Iran*

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Abstract

This paper examines whether the expansion of mobile broadband infrastructure reduces gender gaps in labor-market outcomes in settings characterized by gender-specific employment constraints. Using household survey data from Iran between 2012 and 2020 combined with geocoded 3G coverage data, I exploit the staggered roll-out of 3G infrastructure across counties in a difference-in-differences framework. The results show that 3G expansion increases overall employment and labor force participation by 1.7 and 2.4 percentage points, respectively. These effects are driven primarily by women. Female employment and labor force participation rise by approximately 3 and 4 percentage points, respectively, while no statistically significant effects are found for men. The gains in female employment are not concentrated in part-time or low-earning jobs, suggesting meaningful improvements in labor market engagement. Taken together, the findings indicate that mobile internet expansion can substantially improve labor market outcomes for women in environments where social constraints limit their participation.

Keywords: Mobile broadband; Female employment; Gender gaps

JEL Codes: J16, J21, O33

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1 Introduction

Iran exhibits one of the lowest female employment rates in the world. Despite major socioeconomic changes over the past decade, including a substantial rise in women’s educational attainment and a rapid decline in fertility, the female employment rate has remained close to 10 percent, compared with more than 70 percent for men (Table 1). The existing literature attributes this persistent disparity to restrictive social norms and gender-specific constraints that limit women’s physical mobility, restrict access to socially acceptable occupations, and increase the social costs of market work (Majbouri, 2016, 2019; Taheri et al., 2021; Moghadam, 2009).

This paper examines whether improvements in access to digital infrastructure, in particular mobile broadband, can relax these constraints and improve female labor market outcomes. Internet access may lower barriers to work by enabling home-based employment, facilitating remote job search and work arrangements, and reducing the need for commuting. To the extent that these constraints bind more strongly for women, the expansion of digital access is expected to have larger effects on female employment than on male employment.

To investigate this question, this paper analyzes the labor-market effects of the roll-out of third-generation (3G) mobile internet in Iran, with particular attention to women. This episode provides a unique setting to study the short-run labor-market effects of internet access. As shown in Figure 1, internet access expanded rapidly after the introduction of 3G in 2014, rising from 18 percent to 58 percent within five years; by 2018, more than 80 percent of users accessed the internet via mobile networks.¹ The rapid expansion enables the analysis to isolate short-run responses to improved connectivity, primarily operating through reduced costs of labor-market participation under existing norms rather than changes in the norms themselves.

This paper’s empirical strategy exploits the staggered roll-out of 3G mobile internet

¹Quarterly statistical reports published by Iran’s Ministry of Information and Communications Technology; see Ministry of Information and Communications Technology (ICT) (2018).

coverage across Iranian counties in a difference-in-differences framework. Identification comes from comparing changes in labor-market outcomes within counties before and after the introduction of 3G coverage to contemporaneous changes in counties that had not yet received coverage. The analysis uses rich individual- and household-level data from Iranian Household Expenditure and Income Survey.

To address potential biases arising from staggered treatment timing (Baker et al., 2022), all specifications are estimated using the doubly robust difference-in-differences estimator proposed by Callaway and Sant’Anna (2021) and extended two-way fixed effects estimator suggested by Wooldridge (2025). These estimators accommodates treatment effect heterogeneity across cohorts and over time.

The results indicate that the expansion of 3G network coverage led to statistically significant increases in both the employment rate and labor force participation. In the full sample, 3G rollout raises employment and labor force participation by 1.7 and 2.4 percentage points, respectively.

These effects are driven primarily by women. Female employment rate, and labor force participation rate increases by approximately 3 and 4 percent respectively following the expansion of 3G coverage. This is not only statistically significant, but also economically large in its magnitude relative to a pre-treatment female employment rate and labor force participation of 10 and 13 percent. In contrast, there is no statistically significant effect on male employment or labor force participation.

The results further show no statistically significant impact on income conditional on employment for the full sample or by gender. This pattern suggests that the effects of 3G expansion operate primarily along the extensive margin, increasing labor market participation rather than affecting earnings among the employed. Consistent with this interpretation, the absence of a decline in female labor income implies that newly created jobs following 3G expansion offer wages comparable to the existing female average. Moreover, there is no evidence that these gains are driven by increases in part-time work, self-employment, or low-

earning jobs. Instead, the results indicate that the effects reflect substantive labor market participation in full-time wage and salary employment.

Additional heterogeneity analyses indicate that the gains in female employment are concentrated among unmarried women and those without young children (ages 0–12). These patterns are consistent with the interpretation that childcare-related constraints play an important role in limiting women’s labor market participation.

More broadly, the results suggest that the labor market effects of internet infrastructure expansion depend on the underlying institutional and social environment. While internet access appears to have limited effects on male employment, it can be particularly effective in easing constraints that disproportionately limit women’s labor supply, thereby generating meaningful gains in female employment.

The findings are robust across a wide range of alternative specifications and validation tests. The results remain robust when using alternative continuous measures of 3G coverage. They are also not sensitive to controlling for 2G network, excluding the capital county of Tehran which include one sixth of population and alternative set of control variables. Event-study estimates show no evidence of differential pre-treatment trends, either overall or by gender.

This paper contributes to the growing literature on the labor market effects of internet access (Hjort and Tian, 2021; Hjort and Poulsen, 2019; Bhuller et al., 2023; Viollaz and Winkler, 2022; Dutz et al., 2017), as well as recent studies on mobile broadband expansion (Bahia et al., 2020, 2021; Masaki et al., 2020).

A growing body of the literature shows that improvements in digital connectivity can generate meaningful economic gains. Hjort and Poulsen (2019) exploit the arrival of submarine internet cables across Africa and find substantial employment gains, particularly in higher-skill occupations. Using spatial and temporal variation in broadband rollout, Bahia et al. (2021) and Bahia et al. (2020) estimate the causal effects of mobile broadband expansion in Tanzania and Nigeria, respectively, and show that it increases household consumption

and reduces poverty, with improvements in labor market outcomes as an important channel. Similarly, Masaki et al. (2020) documents positive effects of broadband access on consumption and poverty in Senegal, with larger impacts in urban areas and suggestive evidence that labor market improvements mediate these gains.

A related strand of the literature highlights gender heterogeneity in the labor market effects of internet access. Several studies find that improvements in digital connectivity disproportionately benefit women. For example, Bahia et al. (2021) document larger increases in labor force participation and employment among women following 3G expansion. Viollaz and Winkler (2022) show that internet adoption raises female labor force participation in Jordan, partly through increased online job search. Evidence from Viet Nam (Chun and Tang, 2018) further suggests that ICT adoption increases demand for female and skilled labor, consistent with reductions in gender disparities in labor demand. At the same time, Bahia et al. (2020) find more modest average effects for women in Nigeria, highlighting important heterogeneity across contexts and demographic groups.

One key mechanism emphasized in the literature is improved labor market matching. Using quasi-experimental variation in broadband expansion in Norway, Bhuller et al. (2023) show that internet access increases the speed and quality of job matching, leading to higher re-employment rates and wages. Complementary evidence from the United States (Kuhn and Mansour, 2014) indicates that online job search significantly reduces unemployment durations. Together, these findings suggest that reductions in search frictions are an important channel through which internet access affects labor market outcomes, particularly in settings with initially high frictions.

Overall, the existing literature provides substantial evidence that internet access can improve economic and labor market outcomes, particularly for females, although the magnitude and measures of these effects vary across settings and populations.

This paper contributes to this literature along several dimensions. First, it leverages rich, large-scale individual-level microdata that include detailed measures of employment,

labor force participation, and individual income, allowing for a comprehensive assessment of labor market outcomes. Second, it provides new evidence on pronounced gender heterogeneity in a novel setting, Iran, which is a country characterized by one of the lowest female employment rates in the world and thereby extending the external validity of the literature to environments where social constraints are particularly binding. Third, the institutional context strengthens identification. The rapid expansion of mobile broadband during the study period generates substantial variation in internet access, enabling the estimation of the labor market effects. Finally, the paper relies on a coverage index that reflects users' experienced access rather than solely provider-reported infrastructure. This measure is validated by estimating its effect on self-reported internet use in the data (see Appendix C), providing additional support for the empirical strategy.

The remainder of the paper is organized as follows. Section 2 describes the data sources and the construction of the 3G coverage measures. Section 3 outlines the empirical framework and identification strategy. Section 4 presents the main results and robustness checks. Section 5 concludes and discusses the findings.

2 Data

To estimate the impact of 3G coverage on labor-market outcomes, this paper combines detailed data on mobile network infrastructure from the OpenCellid² (OCI) project with socio-economic information from the Iranian Household Expenditure and Income Survey³ (HEIS) covering the period 2013–2020. This section first explains how the 3G coverage measures are constructed using OCI data and then describes the HEIS microdata used to measure household and individual characteristics.

²See OpenCellID (2026).

³See Statistical Center of Iran (2026).

2.1 Measurement of 3G Coverage

The mobile network coverage measures used in this paper are constructed from data provided by the OpenCellID. OCI is a global crowdsourced database of cellular base stations that aggregates GPS observations of radio signals collected via smartphone applications and processes them to infer the geographic locations of individual cell towers. The database reports information on cellular technologies, along with the date each cell is first observed. This information is used to identify 3G base stations in Iran and to track the timing of their deployment.

The baseline 3G coverage indicator is constructed using user-reported observations of 3G network infrastructure from the OpenCellID (OCI) database. A geographic unit is considered covered once a 3G cell is detected in the OCI data. Given the county–urban/rural resolution of the HEIS, coverage is defined at the county level, such that a county is classified as covered when at least one 3G cell is detected in any city within the county. Although Iranian counties may include multiple cities, this aggregation is unlikely to introduce substantial measurement error, as in the vast majority of cases most of a county’s urban population resides in a single city.⁴

Figure 2 presents the share of counties and the share of the population covered by these coverage measures over time. The expansion of 3G infrastructure occurred primarily between 2014Q4 and 2017Q1, beginning in larger counties, which resulted in a rapid initial increase in the covered population. Coverage growth subsequently slowed as deployment expanded to smaller and less densely populated counties. Figure 3 further presents coverage maps for the years 2013–2016, illustrating the spatial diffusion of 3G mobile internet across counties over time.

This binary measure is simple and intuitive and relies on actual user-reported network observations rather than operators’ coverage claims. However, by abstracting from variation in coverage intensity both within and across counties, it may introduce classical measurement

⁴Based on the administrative divisions in 2013, Iran was composed of 391 counties (Shahrestan).

error, which would bias the estimated coefficients toward zero.

To account for heterogeneity in treatment intensity within counties and variation in coverage levels across counties, the analysis additionally employs a continuous measure of 3G coverage. The continuous measure follows Viollaz and Winkler (2022) approach and approximates coverage intensity by assuming each cell provides service within a 500-meter radius of its recorded location⁵. Coverage intensity is then defined as the share of a county’s urban population residing within these service areas⁶. To reduce sensitivity to small variations in this continuous measure, it is primarily used in binned specifications rather than as a fully continuous regressor. Figure A1 shows the average intensity of 3G coverage over time based on this continuous measure.

To further validate the relevance of the 3G coverage index, I examine its effect on self-reported household internet access using a specification analogous to that employed for labor market outcomes (see Appendix C). The results are statistically significant and indicate that 3G expansion increased household internet access by approximately 5 percentage points.

2.2 Employment Outcomes and Socioeconomic Characteristics

Individual-level socioeconomic information is drawn from the Iranian Household Expenditure and Income Survey (HEIS), a nationally representative survey conducted by the Statistical Center of Iran (SCI). This paper uses HEIS microdata from 2013 to 2020, spanning two years prior to the introduction of 3G coverage and the subsequent expansion period during which coverage approached its peak. The survey covers more than 120,000 individuals annually.

To assess the employment impacts of 3G coverage, I estimate its effects on employment, labor force participation, and income conditional on employment⁷. These outcomes are constructed from individuals’ self-reported survey responses, which directly capture their

⁵OCI does not directly report antenna service areas, and estimating precise coverage would require modeling terrain, antenna characteristics, and local infrastructure, which is beyond the scope of this paper.

⁶Coverage is aggregated to the county level as a population-weighted average of city-level coverage.

⁷All monetary variables are deflated using the Consumer Price Index (CPI) published by the Statistical Center of Iran.

labor-market status. The HEIS data further provide detailed information on individual characteristics, including demographics and education, which are used as control variables.

The sample is restricted to individuals aged 18–64, the segment of the population most actively engaged in the labor market and therefore most responsive to employment shocks. In addition, because the rollout of 3G coverage is concentrated in urban areas, the analysis focuses on urban residents, who are most likely to be directly exposed to the treatment. Table A2 reports the sample size before and after data cleaning.

Table 1 reports baseline characteristics for counties classified as early- or late-treated based on the timing of first exposure to 3G mobile internet coverage. Early-treated counties received 3G between 2014Q4 and 2015Q3, while late-treated counties received 3G between 2015Q4 and 2016Q3. Although early-treated counties have substantially larger average populations than late-treated counties, other baseline characteristics are similar across the two groups.

3 Empirical Strategy

This paper identifies the causal effect of mobile broadband access by exploiting the staggered roll-out of third-generation (3G) mobile-network coverage across Iranian counties in a Difference-in-Differences (DiD) framework. Variation in the timing of coverage provides plausibly exogenous variation in exposure, conditional on fixed effects and controls. The design compares within-county changes in labor-market outcomes before and after the introduction of 3G coverage to contemporaneous changes in counties not yet treated. Identification relies on the assumption that, absent 3G expansion, treated and untreated counties would have followed parallel labor-market trends.

To address potential biases arising from staggered treatment timing under a binary treatment framework (Baker et al., 2022), I re-estimate all specifications using the doubly robust difference-in-differences estimator proposed by Callaway and Sant’Anna (2021) (`csdid`),

which accommodates treatment effect heterogeneity across cohorts and over time. Because `csdid` does not readily accommodate time-varying covariates in this setting, I additionally employ the extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`), which allows for the inclusion of such covariates while retaining flexibility in treatment timing.

The baseline specification is given by Equation 1.

$$Y_{ijt} = \alpha_j + \delta_t + \beta 3G_{jt} + \Gamma X_{it} + \varepsilon_{ijt}, \quad (1)$$

where Y_{ijt} denotes the labor-market outcome of individual i in county j at quarter t , including employment, labor force participation, and labor income. The treatment variable $3G_{jt}$ is a binary indicator equal to one if county j is covered by a 3G network in quarter t . The coefficient of interest, β , captures the average effect of 3G coverage on labor-market outcomes. As treatment varies at the county level, standard errors are clustered at the county level in all regressions.

County fixed effects (α_j) absorb time-invariant differences across counties, while quarter fixed effects (δ_t) capture common aggregate shocks. The vector X_{it} includes a rich set of individual- and household-level controls, including individual demographics (age group, college education, student status, gender, and marital status) and household composition measures, such as the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

To assess the validity of the parallel trends assumption and examine the dynamic effects of 3G coverage, I estimate an event-study specification. In additional specifications, I allow for province-specific linear time trends to account for differential pre-existing trends across regions (see Appendix B), and I further interact 3G coverage with indicators for individual and household characteristics to examine treatment effect heterogeneity.

Finally, I employ an alternative continuous measure of 3G coverage within a binned dose–response framework. This approach captures variation in treatment intensity both

within and across counties. Given potential measurement error in the continuous measure, I discretize it into bins to reduce sensitivity to small fluctuations. To incorporate this measure into the estimators proposed by Callaway and Sant’Anna (2021) and Wooldridge (2025), I define treatment timing as the first quarter in which county-level 3G coverage exceeds a given threshold.

4 Effects of 3G Coverage on Labor Market Outcomes

Table 2 presents estimates of the effect of 3G coverage on employment across alternative specifications and DiD estimators. Panel A reports results for the full sample. Columns 1–3 compare estimates from OLS, `csdid`, and `jwdid` estimators, each including only time fixed effects. The results from `csdid` and `jwdid` estimators are similar to one another but differ from the OLS results.

As discussed in the methodology section, the presence of staggered treatment timing and dynamic treatment effects renders OLS estimates susceptible to bias arising from negative weighting. Given these concerns and the observed differences in estimates, I base the analysis on the `csdid` and `jwdid` approaches. While `csdid` and `jwdid` rely on closely related identification strategies, which is reflected in the similarity of estimates in columns 2 and 3, `csdid` does not accommodate time-varying individual-level covariates in this set-up. Therefore, the preferred specification relies on `jwdid`.

The preferred specification (column 5) shows a large and statistically significant increase in employment associated with the expansion of 3G coverage, even after controlling for individual and household characteristics as well as county fixed effects. Panels B and C replicate the analysis separately for females and males. The results reveal substantial gender heterogeneity: the employment effect is economically and statistically significant for females, with an increase of approximately 3 percentage points, whereas the effect for males is small in magnitude and statistically insignificant. These findings suggest that the primary beneficia-

ries of 3G expansion are women.

Table 3 extends the preferred `jwdid` specification (Column 5 of Table 2) to additional labor market outcomes, including labor force participation and the inverse hyperbolic sine of income conditional on employment. Column (2) shows that 3G coverage increases labor force participation by 3.2 percentage points, with the estimate statistically significant at conventional levels. Consistent with the employment results, the effects display substantial gender heterogeneity. The increase in labor force participation is concentrated among females, for whom participation rises by approximately 3 percentage points, whereas the corresponding effect for males is small in magnitude and statistically indistinguishable from zero.

Column (3) examines the effect of 3G coverage on labor income among employed individuals. The results indicate no statistically significant effect across the full sample as well as in the gender-disaggregated analyses for both males and females. This suggests that the effect of 3G internet operates primarily along the extensive margin rather than through changes in earnings conditional on employment. Notably, the absence of a decline in female labor income indicates that the new jobs created following 3G expansion likely pay wages comparable to the female average.

Figure 4 presents event study estimates based on the `csdid` approach, separately for males and females. There is no evidence of significant pre-trends across groups, supporting the validity of the identification strategy. Following treatment, female employment increases significantly, whereas there is no evidence of a corresponding increase in male employment.

Finally, Table 4 estimates a specification that interacts the treatment indicator with a female dummy. The results confirm the earlier findings: the effect of 3G coverage is not statistically significant for males, while the interaction term is positive and significant, indicating that the impact on female employment is significantly larger than that on male employment.

Appendix B presents a range of robustness checks for the main results on female employment. Tables B6 to B13 shows the findings are robust to alternative sets of covariates,

including specifications that allow for province-specific linear trends across the 30 provinces, and to alternative estimators, including `ols`, `jwddid` and `csddid`. Table B1 shows that the results are not sensitive to excluding the capital city of Tehran, which accounts for approximately one-sixth of the total population. Table B2 demonstrates that the results remain robust when controlling for earlier-generation mobile networks.

4.1 Heterogeneity Among Women

Figure 5 examines heterogeneity in the effects of 3G coverage by individual characteristics. Specifically, I split the sample by gender and estimate separate models for females and males in which the treatment variable is interacted with individual characteristics. Subgroup effects are then recovered from the coefficients on the treatment and interaction terms. Among females, the effects are larger for unmarried individuals and for those without children below the age of twelve. In contrast, the effects do not vary meaningfully by age, education, or the presence of other young household members.

4.2 Effects by Employment Type and Hours

Table 5 breaks down the overall effect on employment by hours worked and employment type. Columns (1) and (2) report the effects on full-time and part-time employment, respectively. The results indicate that the effects are concentrated in full-time employment, with little evidence of an impact on part-time employment. In fact, nearly all of the increase in female employment can be attributed to gains in full-time positions.

Columns (3) and (4) further distinguish between wage and salary employment and self-employment. The results show that the effects are larger for wage employment than for self-employment. In particular, 3G coverage leads to a statistically significant increase of approximately 2 percentage points in wage employment among females, while the estimated effect on self-employment (around 0.7 percentage points) is not statistically significant. These patterns are consistent with the observed increase in female employment being driven by

gains in wage employment rather than self-employment.

4.3 Continuous measure of 3G coverage

Table B14 presents estimates of the effect of 3G coverage on employment using alternative definitions of the treatment variable based on thresholds of a continuous measure of coverage. The results indicate that the largest effects arise when comparing areas with coverage below 10 percent to those with coverage above 10 percent. This suggests that the employment effects of 3G expansion begin to materialize once coverage exceeds a relatively low threshold.

As the threshold used to define treatment increases, the estimated effects remain statistically significant but decline in magnitude. For example, in column 5, where “untreated” areas are defined as those with coverage below 50 percent and “treated” areas as those above 50 percent, the estimated effect is smaller. This pattern is consistent with diminishing marginal effects of 3G coverage on employment.

4.4 Other dimensions

Table B3 reports heterogeneous effects of 3G coverage across additional individual characteristics. The results indicate that the employment effects are larger for individuals without a college education and for married individuals. The effect on employment is larger among individuals older than 35, whereas the effect on labor force participation is larger for those below age 34. This suggests that, for younger individuals, the impact of 3G expansion operates more through changes in unemployment (i.e., entry into the labor force) rather than transitions into employment.

Table B4 presents a similar analysis based on household characteristics, such as the presence of a young adult or a child below age 12. The results show that the effects are larger for individuals living in households with young members, particularly those with children.

5 Conclusion

Iran exhibits one of the lowest female employment rates in the world, with the female employment rate remaining near 10 percent despite substantial gains in women’s educational attainment and sharp declines in fertility over the past decade. This paper exploits the staggered rollout of 3G mobile coverage across counties in a difference-in-differences framework to study whether improved access to mobile broadband can mitigate gender disparities in labor-market outcomes.

The results indicate that 3G internet expansion increased employment, and labor force participation. Importantly, results display substantial heterogeneity by gender. Despite a statistically insignificant average effect on males, female employment and labor force participation rise by approximately 3 and 4 percentage points, respectively a sizable effect compared to a pre-treatment female employment rate of roughly 10 and 13 percent. These gains are concentrated primarily among unmarried woman and woman without a kid below 12 years old.

The timing of the female employment response provides further insight. The effect of 3G coverage on female employment emerges within less than a year. However, the social norms that largely underlie low female employment are unlikely to change over such a short period. This suggests that expanded internet access may enable women’s labor-market participation by lowering the cost of complying with existing norms, without necessarily changing the norms themselves.

Finally, the composition of employment gains suggests that the additional jobs held by women are not marginal or peripheral positions, but reflect meaningful attachment to the labor market. The increase in female employment is not concentrated in part-time positions, and average earnings among employed women do not decline following the expansion of 3G coverage. Moreover, the gains are concentrated in wage and salary employment rather than self-employment. While these patterns do not fully capture all dimensions of job quality, they are consistent with the interpretation that the observed employment increases reflect

substantive labor market participation rather than part-time or low-income self-employment.

These findings complement the growing literature on internet adoption and labor markets by showing that, depending on the institutional and social context, the same technological change may have limited average effects, yet generate meaningful distributional consequences by easing the constraints faced by disadvantaged groups

Tables

Table 1: Baseline Characteristics of Counties by Timing of 3G Rollout

	Early-treated	Late-treated
<i>Panel A. Demographics and human capital</i>		
Female (%)	50.61	51.40
Young (15–34) (%)	32.77	31.38
Age 18–64 (%)	68.32	64.83
College degree (%)	14.77	10.42
Student (%)	19.71	19.86
Married (%)	53.53	53.98
Internet access (%)	20.20	9.19
Mobile phone ownership (%)	89.33	84.88
Computer ownership (%)	34.02	23.13
<i>Panel B. Labor market outcomes</i>		
Economically active population (%)	68.32	64.83
Employed (%)	27.12	26.93
Unemployed (%)	4.40	4.73
Labor force participation (%)	31.53	31.67
<i>Panel C. Household composition</i>		
Young adult member (15–34) (%)	75.86	75.13
College-educated member (%)	33.81	27.35
Student member (%)	52.07	52.98
Household size	3.45	3.56
<i>Panel D. County characteristics</i>		
Number of counties	135	198
Total population	2,570,000	139,712
Urban population	2,390,000	80,070
Manufacturing employment share	17.66	12.09

Notes: The table reports baseline characteristics for counties classified as early-treated (2014Q4–2015Q3) or late-treated (2015Q4–2016Q3) based on the timing of first exposure to 3G mobile internet coverage. Household- and individual-level statistics (Panels A–C) are computed using pre-treatment data from the Iranian Household Expenditure and Income Survey (HEIS), 2013Q2–2014Q3, and are weighted using HEIS sampling weights. Panel D reports county-level characteristics from the 2011 Population Census; county-level variables are unweighted.

Table 2: Effect on 3G coverage on Employment Across Estimators and Specifications

	Baseline Specification			County FE + Controls	
	(1) OLS	(2) <i>csdid</i>	(3) <i>jwddid</i>	(4) OLS	(5) <i>jwddid</i>
<i>Panel A. All Individuals</i>					
Effect	-0.0058 (0.0098)	0.0288*** (0.0066)	0.0269* (0.0143)	-0.0004 (0.0055)	0.0175** (0.0088)
<i>Panel B. Females</i>					
Effect	0.0175 (0.0130)	0.0835*** (0.0107)	0.0620** (0.0302)	0.0057 (0.0045)	0.0301** (0.0151)
<i>Panel C. Males</i>					
Effect	-0.0341*** (0.0118)	-0.0080 (0.0096)	-0.0166 (0.0173)	-0.0061 (0.0087)	0.0020 (0.0151)
Time FE	Yes	Yes	Yes	Yes	Yes
County FE	No	No	No	Yes	Yes
Individual controls	No	No	No	Yes	Yes

Notes: Each entry reports estimates of the effect of 3G coverage on an indicator for employment based on Equation 1. Columns (1) and (4) report OLS estimates. Column (2) reports the simple ATT using doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021) (*csdid*), while Columns (3) and (5) report simple ATT estimates using extended two-way fixed effects estimator of Wooldridge (2025) (*jwddid*). Columns (1)–(3) present baseline specifications, while Columns (4) and (5) additionally include county fixed effects and individual controls. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively.

Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effect of 3G Coverage on Employment Outcomes: *jwddid* Estimates

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. All Individuals</i>			
ATT	0.0175** (0.0088)	0.0237*** (0.0090)	-0.0109 (0.0892)
<i>Panel B. Females</i>			
ATT	0.0301** (0.0151)	0.0399*** (0.0153)	0.2130 (0.5552)
<i>Panel C. Males</i>			
ATT	0.0020 (0.0151)	0.0023 (0.0119)	-0.0984 (0.1010)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes: Each cell reports simple ATT estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using extended two-way fixed effects estimator of Wooldridge (2025) (*jwddid*). Column (1) uses an employment indicator; Column (2) a labor force participation indicator; Column (3) the inverse hyperbolic sine of labor income.

Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively.

Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effect of 3G Coverage by Gender

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. Regression Coefficients</i>			
3G (Effect for Males)	0.0040 (0.0092)	0.0118 (0.0097)	0.0712 (0.0463)
3G $\times$ Female	0.0260*** (0.0043)	0.0227*** (0.0051)	-0.0038 (0.0271)
<i>Panel B. Computed Effect for Females</i>			
3G + 3G $\times$ Female	0.0300*** (0.0090)	0.0345*** (0.0092)	0.0674 (0.0509)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes: Each column reports estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`) with interactions between the treatment variable and a female indicator. Panel A reports the coefficient on 3G (effect for males) and the interaction term (difference between females and males). Panel B reports the implied effect for females, computed as the sum of these coefficients. Column (1) uses an employment indicator; Column (2) a labor force participation indicator; Column (3) the inverse hyperbolic sine of labor income. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of 3G Coverage by Employment Type and Hours: *jwddid* Estimates

	Employment Type		Hours Worked	
	(1) Employee	(2) Self-employed	(3) Full-time	(4) Part-time
<i>Panel A. All Individuals</i>				
ATT	0.0092 (0.0155)	0.0047 (0.0140)	0.0158 (0.0112)	-0.0030 (0.0192)
<i>Panel B. Females</i>				
ATT	0.0235* (0.0131)	0.0075 (0.0068)	0.0320** (0.0139)	0.0017 (0.0095)
<i>Panel C. Males</i>				
ATT	-0.0074 (0.0286)	0.0001 (0.0304)	-0.0030 (0.0192)	-0.0019 (0.0075)
Time FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes

Notes: Each column reports estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using extended two-way fixed effects estimator of Wooldridge (2025) (*jwddid*). Columns (1) and (2) distinguish between wage employment (employee) and self-employment, while Columns (3) and (4) distinguish between full-time and part-time employment. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively.

Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figures

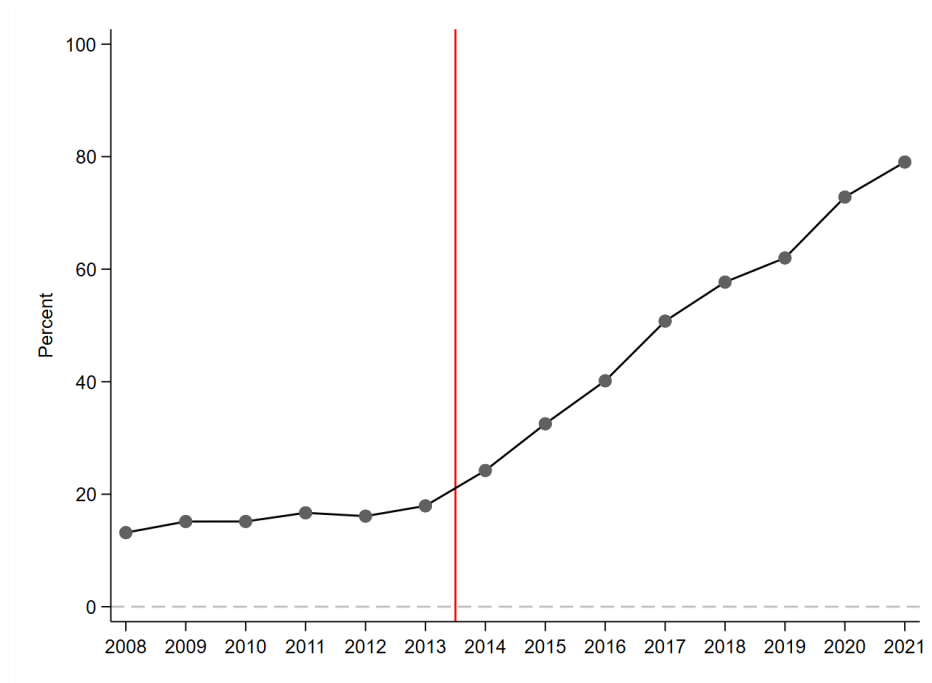
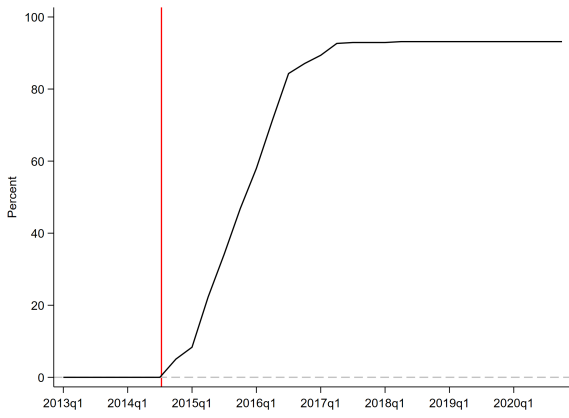
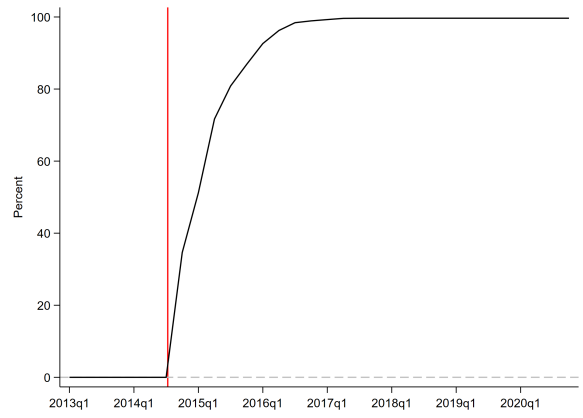


Figure 1: Trends in Household Internet Access in Urban Iran

Notes: The figure plots the share of urban households reporting access to the internet, using data from the Iranian Household Expenditure and Income Survey (HEIS). The vertical red line marks 2014Q4, when the two major mobile network operators began offering 3G mobile internet services. Internet access includes all modes of connection, including mobile and fixed-line services. The sample period spans 2008Q2–2020Q3 (Iranian calendar years 1387–1399).



(a) Share of Counties Covered by 3G



(b) Share of Counties Covered by 3G

Figure 2: Expansion of 3G Mobile Network Coverage in Iran

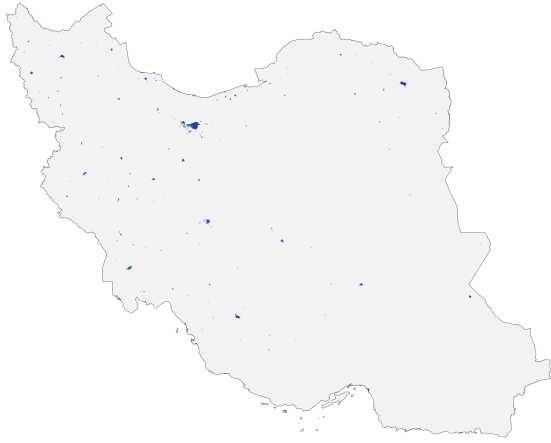
Notes: The figure plots the share of counties and the share of the population covered by binary coverage measures over time using data from OpenCellID. The vertical red line marks 2014Q4, when the two major mobile network operators began offering 3G mobile internet services. The expansion of 3G infrastructure occurred primarily between 2014Q4 and 2017Q1, beginning in larger counties, which resulted in a rapid initial increase in the covered population. Coverage growth subsequently slowed as deployment expanded to smaller and less densely populated counties. The sample is restricted to the two major mobile network operators, MCI (Hamrah Aval) and MTN Irancell



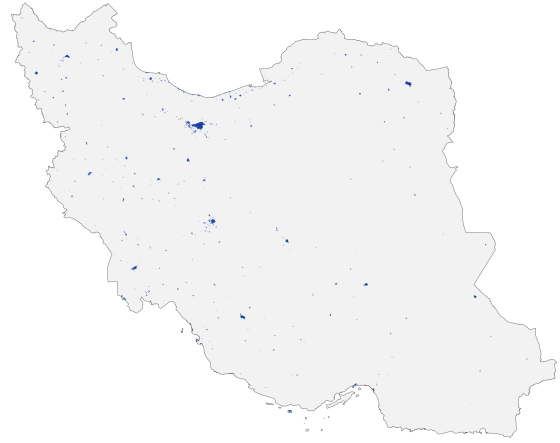
(a) 2013



(b) 2014



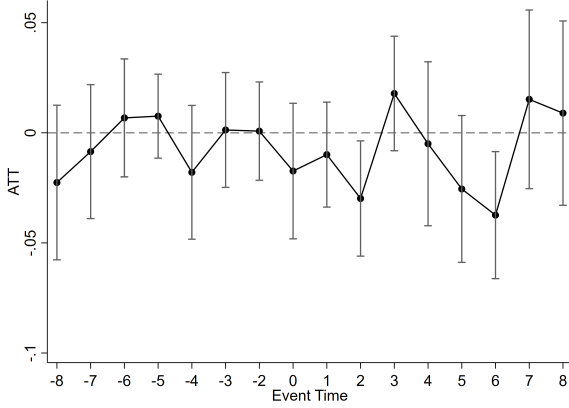
(c) 2015



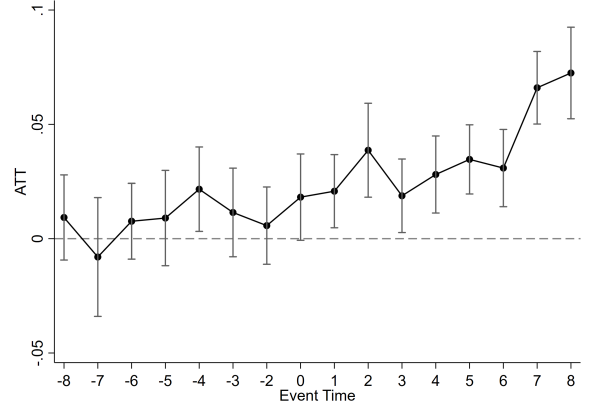
(d) 2016

Figure 3: Geographic Expansion of 3G Coverage

Notes: The maps display the geographic distribution of 3G (UMTS) mobile network cells in Iran using data from OpenCellID. Each cell is assumed to cover a radius of 500 meters. The sample is restricted to the two major mobile network operators, MCI (Hamrah Aval) and MTN Irancell.



(a) Males



(b) Females

Figure 4: `csdid` Event-Study Estimates of the Effect of 3G Coverage on Employment

Notes: Figure shows event study estimation of the effect of 3G coverage on an employment indicator based on Equation 1, estimated using doubly robust difference-in-differences estimator of Callaway and Sant'Anna (2021) (`csdid`). Panels A and B report results for sub sample of females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

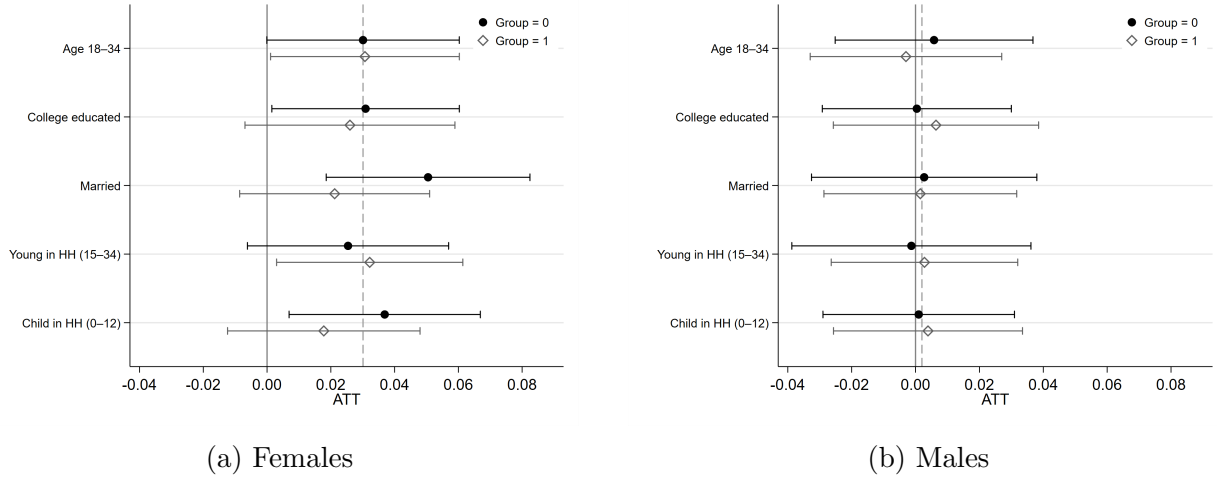


Figure 5: Heterogeneous Effects of 3G Coverage on Employment by Gender: `jwddid` Estimates

Notes: The graph reports estimates from regressions of an employment indicator on a 3G coverage indicator, based on Equation 1 and estimated using extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`), where the treatment variable is interacted with individual or household characteristics. Panels A and B correspond to females and males, respectively. Group = 0 denotes the estimated coefficient when the indicator for the given characteristic equals 0, and Group = 1 when it equals 1. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+). All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

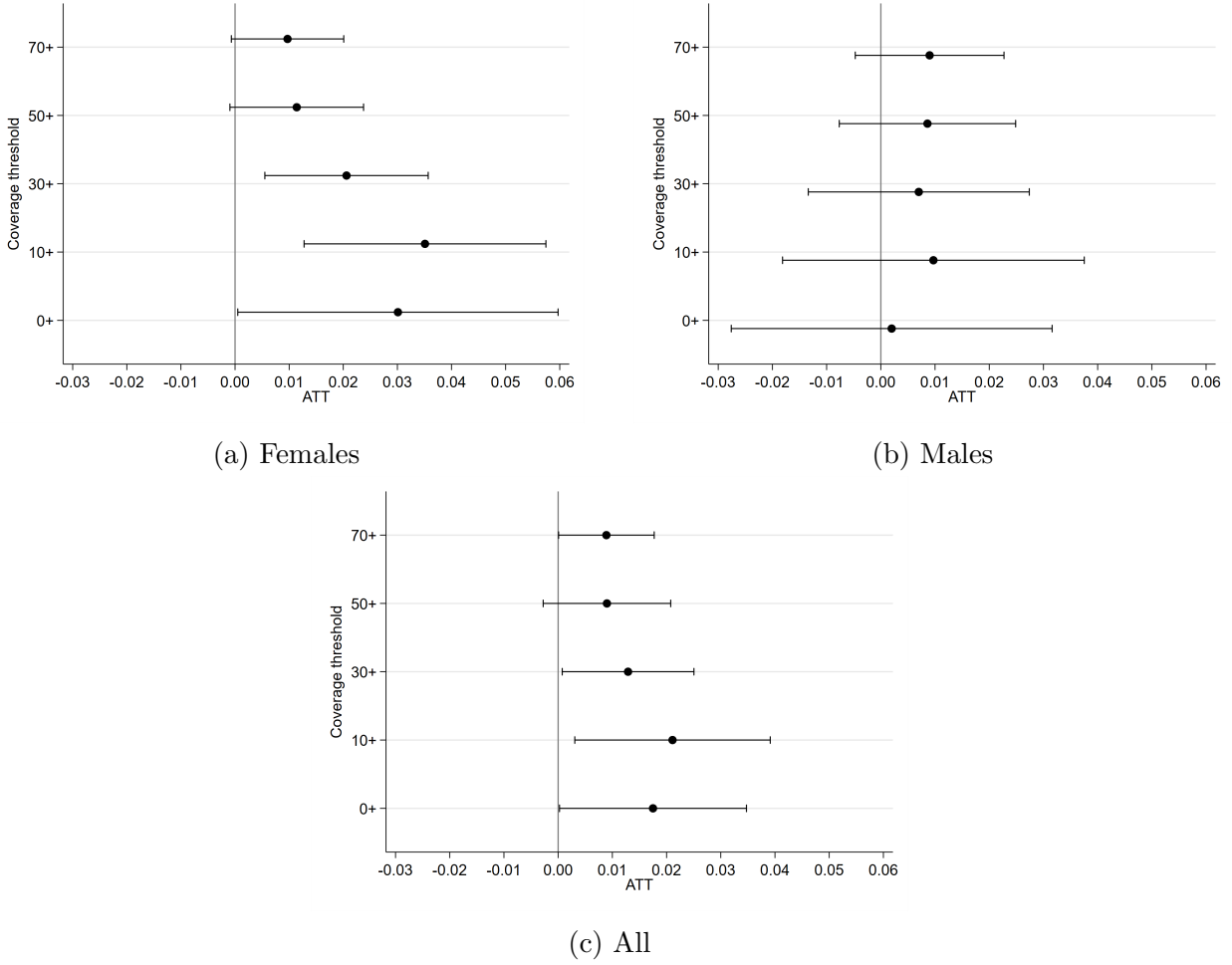


Figure 6: Effect of 3G Coverage on Employment by Coverage Threshold: `jwddid` Estimates

Notes: Graph reports separate estimations using extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`) based on Equation 1 of an indicator for employment on a 3G coverage indicator. Treatment is defined as the first quarter in which county-level coverage exceeds the indicated threshold. Thus, the estimate in each column compares counties that have reached the threshold to counties that remain below that threshold. Panels A and B report results for females and males, respectively, while Panel C reports estimates for the full sample.

Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

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Appendix

A Additional Descriptive Statistics

Table A1: Baseline Socioeconomic Characteristics by Gender

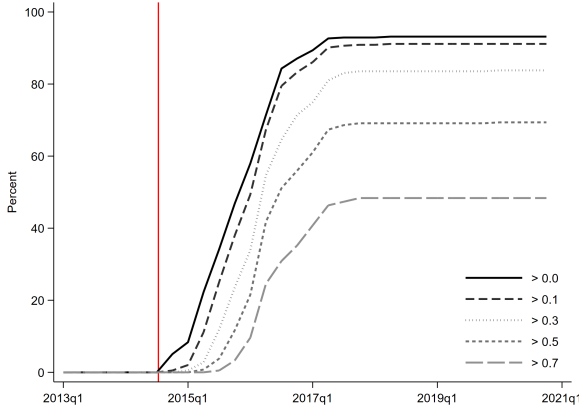
	Male (%)	Female (%)
<i>Panel A. Outcomes and Characteristics</i>		
Employed	69.2	9.8
Unemployed	9.0	3.7
Labor force participation	78.2	13.4
Young (ages 18–34)	49.2	46.9
College degree	21.1	19.0
<i>Panel B. Job Type Composition (Employed Sample)</i>		
Wage and salary workers	65.8	75.9
Self-employed	34.9	24.1
<i>Panel C. Employment Intensity (Employed Sample)</i>		
Part-time employment	10.4	39.4
Full-time employment	89.6	60.6

Notes: The table reports baseline socioeconomic characteristics by gender using data from the Iranian Household Expenditure and Income Survey (HEIS), 2013Q2–2014Q3. All statistics are expressed as percentages and are computed using sampling weights. Panel A reports means for the full sample by gender. Panels B and C report shares computed among employed individuals only (i.e., conditional on employment).

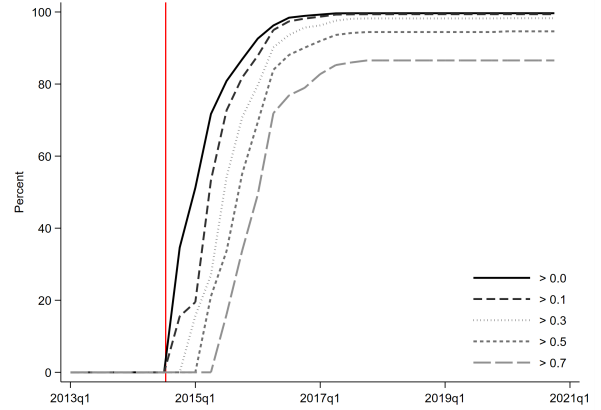
Table A2: Sample Size and Composition Before and After Sample Restrictions

	All individuals	Age 18–64	Urban
<i>Panel A. Total</i>			
Total	1,084,289	683,212	351,275
<i>Panel B. Employment Status</i>			
Employed	304,037	282,114	134,747
Unemployed	56,098	52,847	29,745
<i>Panel C. Gender</i>			
Female	542,310	347,456	177,718
Male	541,979	335,756	173,557

Notes: The table reports the number of observations across subsamples. Data are drawn from the Iranian Household Expenditure and Income Survey (HEIS), 2013Q2–2020Q3 (Iranian calendar years 1392–1399). The column “Age 18–64” restricts the sample to individuals of working age. The column “Urban” further restricts the sample to individuals residing in urban areas.



(a) County-Level Average 3G Coverage Intensity



(b) Population-Weighted Average 3G Coverage Intensity

Figure A1: Evolution of 3G Coverage Intensity Over Time

Notes: The figure shows the average intensity of 3G coverage by counties and average intensity of 3G coverage by population over time using data from OpenCellID. The vertical red line marks 2014Q4, when the two major mobile network operators began offering 3G mobile internet services. The expansion of 3G infrastructure occurred primarily between 2014Q4 and 2017Q1, beginning in larger counties, which resulted in a rapid initial increase in the covered population. Coverage growth subsequently slowed as deployment expanded to smaller and less densely populated counties. The sample is restricted to the two major mobile network operators, MCI (Hamrah Aval) and MTN Irancell

B Effect of 3G on Employment Outcomes: Robustness Checks

Table B1: Effect of 3G Coverage on Employment Outcomes: `jwddid` Estimates, Excluding Tehran

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. All Individuals</i>			
3G	0.0170* (0.0090)	0.0208** (0.0091)	0.0071 (0.0927)
<i>Panel B. Females</i>			
3G	0.0289* (0.0154)	0.0371** (0.0155)	0.3642 (0.5375)
<i>Panel C. Males</i>			
3G	0.0036 (0.0154)	0.0010 (0.0120)	-0.1004 (0.1027)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes: Each cell reports simple ATT estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using the extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`). Column (1) uses an employment indicator; Column (2) a labor force participation indicator; Column (3) the inverse hyperbolic sine of labor income.

Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B2: Effect of 3G Coverage on Employment Outcomes: `jwddid` Estimates Controlling for 2G Coverage

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. All Individuals</i>			
3G	0.0184** (0.0091)	0.0227** (0.0095)	-0.0151 (0.0989)
<i>Panel B. Females</i>			
3G	0.0274* (0.0155)	0.0357** (0.0160)	0.3179 (0.5996)
<i>Panel C. Males</i>			
3G	0.0070 (0.0156)	0.0044 (0.0124)	-0.1129 (0.1030)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
2G coverage control	Yes	Yes	Yes

Notes: Each cell reports simple ATT estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using the extended two-way fixed effects estimator of Wooldridge (2025) (`jwddid`). Column (1) uses an employment indicator; Column (2) a labor force participation indicator; Column (3) the inverse hyperbolic sine of labor income.

Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3: Heterogeneous Effects of 3G Coverage by Demographic Characteristics: *jwddid* Estimates

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. Education</i>			
College degree	-0.0265 (0.0252)	-0.0210 (0.0238)	0.1059 (0.0971)
No college degree	0.0221** (0.0087)	0.0245*** (0.0087)	0.0691 (0.0480)
<i>Panel B. Age</i>			
Age 18–34	0.0111 (0.0124)	0.0266** (0.0112)	0.0771 (0.0509)
Age 35+	0.0228* (0.0124)	0.0197 (0.0127)	0.0760 (0.0671)
<i>Panel C. Marital Status</i>			
Married	0.0266** (0.0109)	0.0265** (0.0104)	0.0725* (0.0394)
Not married	-0.0306 (0.0243)	-0.0108 (0.0248)	-0.0680 (0.1005)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes: Each column reports estimates of the effect of 3G coverage on the indicated labor-market outcome for the specified subgroup, based on Equation 1 and estimated using the extended two-way fixed effects estimator of Wooldridge (2025) (*jwddid*) Panels correspond to different dimensions of heterogeneity: education, age, and marital status.

All specifications include county and time fixed effects, as well as the full set of individual controls. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4: Heterogeneous Effects of 3G Coverage by Household Composition: *jwddid* Estimates

	(1) Employment	(2) LFP	(3) Income
<i>Panel B. Household Member Age 15–34</i>			
Has member age 15–34	0.0219** (0.0109)	0.0301*** (0.0105)	0.0611 (0.0409)
No member age 15–34	0.0009 (0.0253)	-0.0029 (0.0225)	0.1487 (0.1290)
<i>Panel C. Household Member Age 0–12</i>			
Has member age 0–12	0.0311*** (0.0117)	0.0315** (0.0130)	0.0256 (0.0428)
No member age 0–12	0.0031 (0.0138)	0.0176 (0.0145)	0.1113* (0.0652)
Time FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes: Each column reports estimates of the effect of 3G coverage on the indicated labor-market outcome for the specified subgroup, based on Equation 1 and estimated using the extended two-way fixed effects estimator of Wooldridge (2025) (*jwddid*). Panel A reports estimates for the full sample. Panels B and C report results by household composition, based on the presence of members aged 15–34 and children aged 0–12, respectively.

All specifications include county and time fixed effects, as well as the full set of individual controls. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: Heterogeneous Effects of 3G Coverage on Employment by Presence of Children:
jwddid Estimates

	Female	Male
<i>Panel A. Children Ages 0–2</i>		
Has children	0.0071 (0.0154)	-0.0046 (0.0155)
No children	0.0325** (0.0152)	0.0026 (0.0151)
<i>Panel B. Children Ages 3–5</i>		
Has children	0.0228 (0.0157)	0.0109 (0.0154)
No children	0.0311** (0.0151)	0.0008 (0.0151)
<i>Panel C. Children Ages 6–12</i>		
Has children	0.0220 (0.0152)	-0.0001 (0.0152)
No children	0.0328** (0.0152)	0.0026 (0.0152)
<i>Panel D. Children Ages 0–6</i>		
Has children	0.0098 (0.0158)	0.0084 (0.0151)
No children	0.0357** (0.0152)	0.0002 (0.0152)
Time FE	Yes	Yes
County FE	Yes	Yes
Controls	Yes	Yes

Notes: Each entry reports jwddid estimates of the effect of 3G coverage on an employment indicator for the specified subgroup, based on Equation 1. Panels correspond to different age ranges of children present in the household. All specifications include county and time fixed effects, as well as a full set of individual-level controls. Standard errors clustered at the county level are reported in parentheses. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.1 jwddid Estimations

Table B6: Effect of 3G Coverage on Employment: jwddid Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	0.0269* (0.0143)	0.0201** (0.0089)	0.0175** (0.0088)	0.0183** (0.0092)
<i>Panel B. Females</i>				
3G	0.0620** (0.0302)	0.0400** (0.0167)	0.0301** (0.0151)	0.0373** (0.0152)
<i>Panel C. Males</i>				
3G	-0.0166 (0.0173)	-0.0033 (0.0164)	0.0020 (0.0151)	-0.0038 (0.0184)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports jwddid estimates of the effect of 3G coverage on an employment indicator based on Equation 1. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B7: Effect of 3G Coverage on Labor Force Participation: *jwddid* Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	0.0408** (0.0170)	0.0307*** (0.0082)	0.0237*** (0.0090)	0.0231** (0.0093)
<i>Panel B. Females</i>				
3G	0.0766** (0.0311)	0.0549*** (0.0161)	0.0399*** (0.0153)	0.0463*** (0.0159)
<i>Panel C. Males</i>				
3G	-0.0045 (0.0149)	0.0026 (0.0136)	0.0023 (0.0119)	-0.0054 (0.0127)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports *jwddid* estimates of the effect of 3G coverage on an labor force participation indicator based on Equation 1. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B8: Effect of 3G Coverage on inverse hyperbolic sine of labor income: `jwdid` Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	-0.0919 (0.1707)	-0.0176 (0.0846)	-0.0109 (0.0892)	-0.0127 (0.0875)
<i>Panel B. Females</i>				
3G	0.9579* (0.4951)	0.2310 (0.6142)	0.2130 (0.5552)	0.2270 (0.5569)
<i>Panel C. Males</i>				
3G	-0.1947 (0.2153)	-0.0795 (0.1065)	-0.0984 (0.1010)	-0.0868 (0.0962)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports `jwdid` estimates of the effect of 3G coverage on on inverse hyperbolic sine of labor income, conditional on employment, based on Equation 1. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 csdid Estimations

Table B9: Effect of 3G Coverage on Employment Outcomes: `csdid` Estimates

	(1) Employment	(2) LFP	(3) Income
<i>Panel A. All Individuals</i>			
ATT	0.0288*** (0.0066)	0.0323*** (0.0073)	0.0573 (0.0830)
<i>Panel B. Females</i>			
ATT	0.0835*** (0.0107)	0.0929*** (0.0114)	0.3011* (0.1812)
<i>Panel C. Males</i>			
ATT	-0.0080 (0.0096)	-0.0120 (0.0092)	0.0510 (0.1091)
Time FE	Yes	Yes	Yes
County FE	No	No	No
Individual controls	No	No	No

Notes: Each cell reports simple ATT estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1, estimated using the doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021) (`csdid`). Column (1) uses an employment indicator; Column (2) a labor force participation indicator; Column (3) the inverse hyperbolic sine of labor income.

Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B10: Effect of 3G Coverage on Labor Market Outcomes: `csdid` Estimates (with Province Trends)

	Employment (1)	LFP (2)	Income (3)
<i>Panel A. All Individuals</i>			
3G	0.0277** (0.0141)	0.0296** (0.0146)	0.0788 (0.1149)
<i>Panel B. Females</i>			
3G	0.0703** (0.0291)	0.0770*** (0.0269)	0.9869*** (0.3651)
<i>Panel C. Males</i>			
3G	-0.0005 (0.0225)	-0.0067 (0.0232)	-0.0286 (0.1422)
Time FE	Yes	Yes	Yes
Individual controls	No	No	No
Province $\times$ Trend	Yes	Yes	Yes

Notes: Each entry reports `csdid` estimates of the effect of 3G coverage on labor market outcomes based on Equation 1. Column (1) reports effects on employment, column (2) on labor force participation, and column (3) on inverse hyperbolic sine of labor income (conditional on employment). All specifications include time fixed effects and province-specific linear trends. Panels A–C report results for the full sample, females, and males, respectively. Standard errors clustered at the location level are reported in parentheses. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 `ols` Estimation

Table B11: Effect of 3G Coverage on Employment: `ols` Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	-0.0058 (0.0098)	-0.0067 (0.0063)	-0.0004 (0.0055)	0.0020 (0.0051)
<i>Panel B. Females</i>				
3G	0.0175 (0.0130)	0.0029 (0.0051)	0.0057 (0.0045)	0.0099** (0.0046)
<i>Panel C. Males</i>				
3G	-0.0341*** (0.0118)	-0.0194* (0.0101)	-0.0061 (0.0087)	-0.0049 (0.0089)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports OLS estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1. The dependent variable is an employment indicator. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B12: Effect of 3G Coverage on Labor Force Participation: *ols* Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	-0.0064 (0.0077)	-0.0033 (0.0052)	0.0021 (0.0043)	0.0042 (0.0040)
<i>Panel B. Females</i>				
3G	0.0193 (0.0140)	0.0040 (0.0060)	0.0086* (0.0049)	0.0118** (0.0050)
<i>Panel C. Males</i>				
3G	-0.0375*** (0.0074)	-0.0135** (0.0068)	-0.0035 (0.0062)	-0.0023 (0.0062)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports OLS estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1. The dependent variable is a indicator for labor force participation. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B13: Effect of 3G Coverage on inverse hyperbolic sine of labor income: `ols` Estimates

	(1)	(2)	(3)	(4)
<i>Panel A. All Individuals</i>				
3G	0.4029*** (0.0958)	-0.0516 (0.0429)	-0.0198 (0.0406)	-0.0275 (0.0427)
<i>Panel B. Females</i>				
3G	1.1712*** (0.3206)	0.0462 (0.1924)	0.1848 (0.1720)	0.1037 (0.2172)
<i>Panel C. Males</i>				
3G	0.3124*** (0.0740)	-0.0565 (0.0371)	-0.0351 (0.0381)	-0.0375 (0.0353)
Time FE	Yes	Yes	Yes	Yes
County FE	No	Yes	Yes	Yes
Individual controls	No	No	Yes	Yes
Province $\times$ Trend	No	No	No	Yes

Notes: Each entry reports OLS estimates of the effect of 3G coverage on the indicated labor-market outcome based on Equation 1. The dependent variable is the inverse hyperbolic sine of labor income. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Continuous Measure of 3G coverage

Table B14: Effect of 3G Coverage on Employment by Coverage Threshold: jwddid Estimates

	<i>3G Coverage Threshold</i>				
	(1) 0+	(2) 10+	(3) 30+	(4) 50+	(5) 70+
<i>Panel A. All Individuals</i>					
ATT	0.0175** (0.0088)	0.0211** (0.0092)	0.0129** (0.0062)	0.0090 (0.0060)	0.0089** (0.0045)
<i>Panel B. Females</i>					
ATT	0.0301** (0.0151)	0.0351*** (0.0114)	0.0206*** (0.0077)	0.0114* (0.0063)	0.0097* (0.0053)
<i>Panel C. Males</i>					
ATT	0.0020 (0.0151)	0.0097 (0.0142)	0.0070 (0.0104)	0.0086 (0.0083)	0.0090 (0.0070)
Time FE	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes

Notes: Each column reports estimates of the effect of 3G coverage on an employment indicator based on Equation 1, estimated using the extended two-way fixed effects estimator of Wooldridge (2025) (jwddid). Treatment is defined as the first quarter in which county-level 3G coverage exceeds the indicated threshold. Each column therefore compares counties that have reached the threshold to those that remain below it. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively.

All specifications include county and time fixed effects, as well as the full set of individual controls. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B15: Effect of 3G Coverage on Employment by Coverage Threshold: *csdid* Estimates

	<i>3G Coverage Threshold</i>				
	(1) 0+	(2) 10+	(3) 30+	(4) 50+	(5) 70+
<i>Panel A. All Individuals</i>					
3G	0.0288*** (0.0066)	0.0204** (0.0096)	0.0192*** (0.0057)	0.0165*** (0.0057)	0.0098 (0.0089)
<i>Panel B. Female</i>					
3G	0.0835*** (0.0107)	0.0349** (0.0148)	0.0084 (0.0063)	-0.0014 (0.0075)	-0.0088 (0.0069)
<i>Panel C. Male</i>					
3G	-0.0080 (0.0096)	0.0047 (0.0137)	0.0312*** (0.0101)	0.0339*** (0.0081)	0.0255 (0.0205)
Time FE	Yes	Yes	Yes	Yes	Yes
County FE	No	No	No	No	No
Individual controls	No	No	No	No	No

Notes: Each column reports estimates of the effect of 3G coverage on an employment indicator based on Equation 1, estimated using the doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021) (*csdid*). Treatment is defined as the first quarter in which county-level 3G coverage exceeds the indicated threshold. Each column therefore compares counties that have reached the threshold to those that remain below it. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. All specifications include county and time fixed effects, as well as the full set of individual controls. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Effect of 3G Coverage on Self-Reported Internet Access

Table C1: Effect of 3G Coverage on Self-Reported Internet Access

	Baseline Specification			Location FE + Controls	
	(1) OLS	(2) CSDID	(3) JWDID	(4) OLS	(5) JWDID
<i>Panel A. All Individuals</i>					
Effect	0.1421*** (0.0423)	0.0279** (0.0112)	0.0393 (0.0247)	0.0209 (0.0157)	0.0479** (0.0235)
Time FE	Yes	Yes	Yes	Yes	Yes
Location FE	No	No	No	Yes	Yes
Individual controls	No	No	No	Yes	Yes

Notes: Each entry reports estimates of the effect of 3G coverage on an indicator for self-reported internet access in home, based on Equation 1. Columns (1) and (4) report OLS estimates. Column (2) reports the simple ATT using `csdid`, while Columns (3) and (5) report simple ATT estimates using the extended two-way fixed effects estimator of Wooldridge (2025) (`jwdid`). Columns (1)–(3) present baseline specifications, while Columns (4) and (5) additionally include county fixed effects and individual controls. Panel A reports estimates for the full sample, while Panels B and C report results for females and males, respectively. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

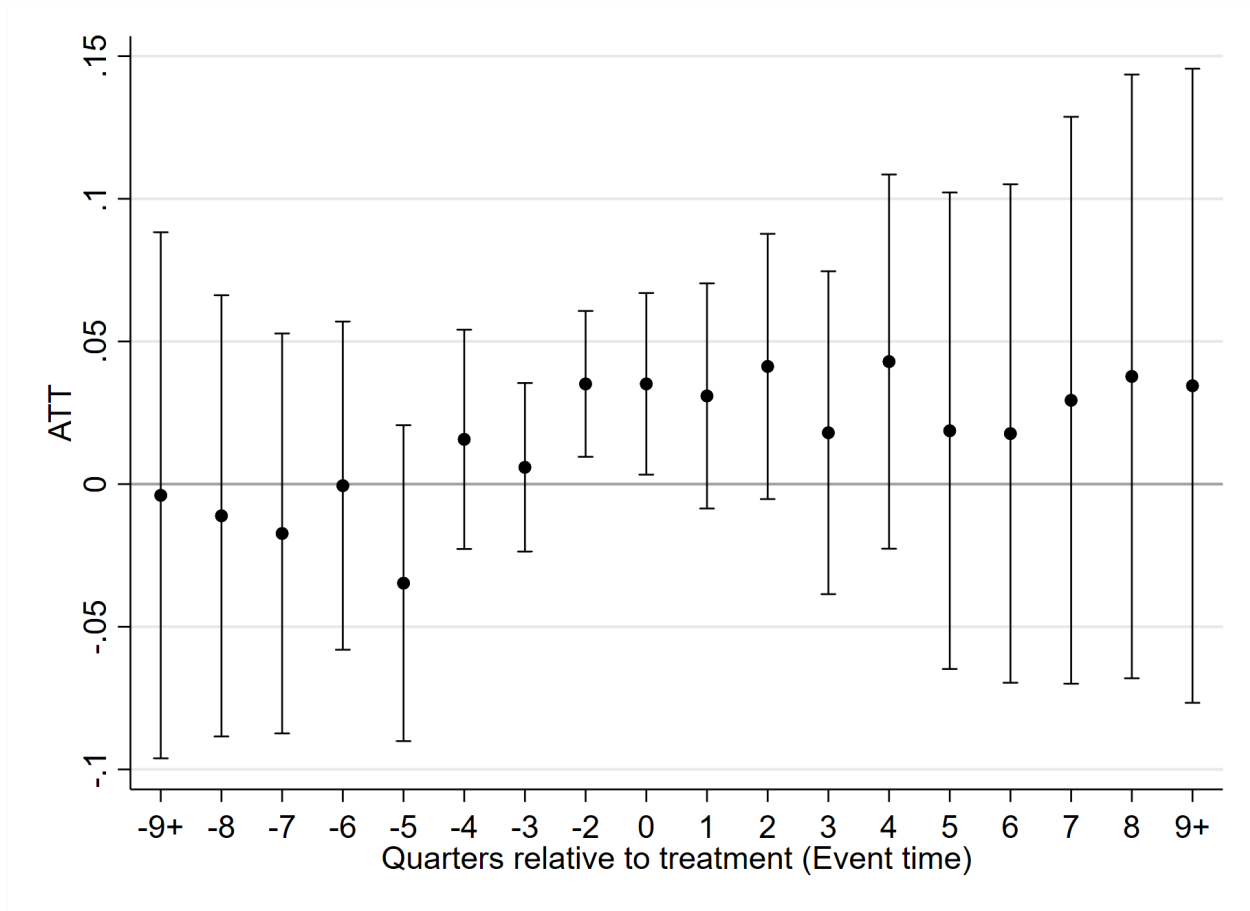


Figure C1: Event Study Estimates of the Effect of 3G Coverage on Self-Reported Internet Access *ols* Estimator

Notes: The figure reports event-study estimates of the effect of 3G coverage on an indicator for self-reported internet access at home, based on Equation 1 and estimated using an OLS specification. The coefficients trace the dynamic effects relative to the period prior to the introduction of 3G coverage. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+). All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

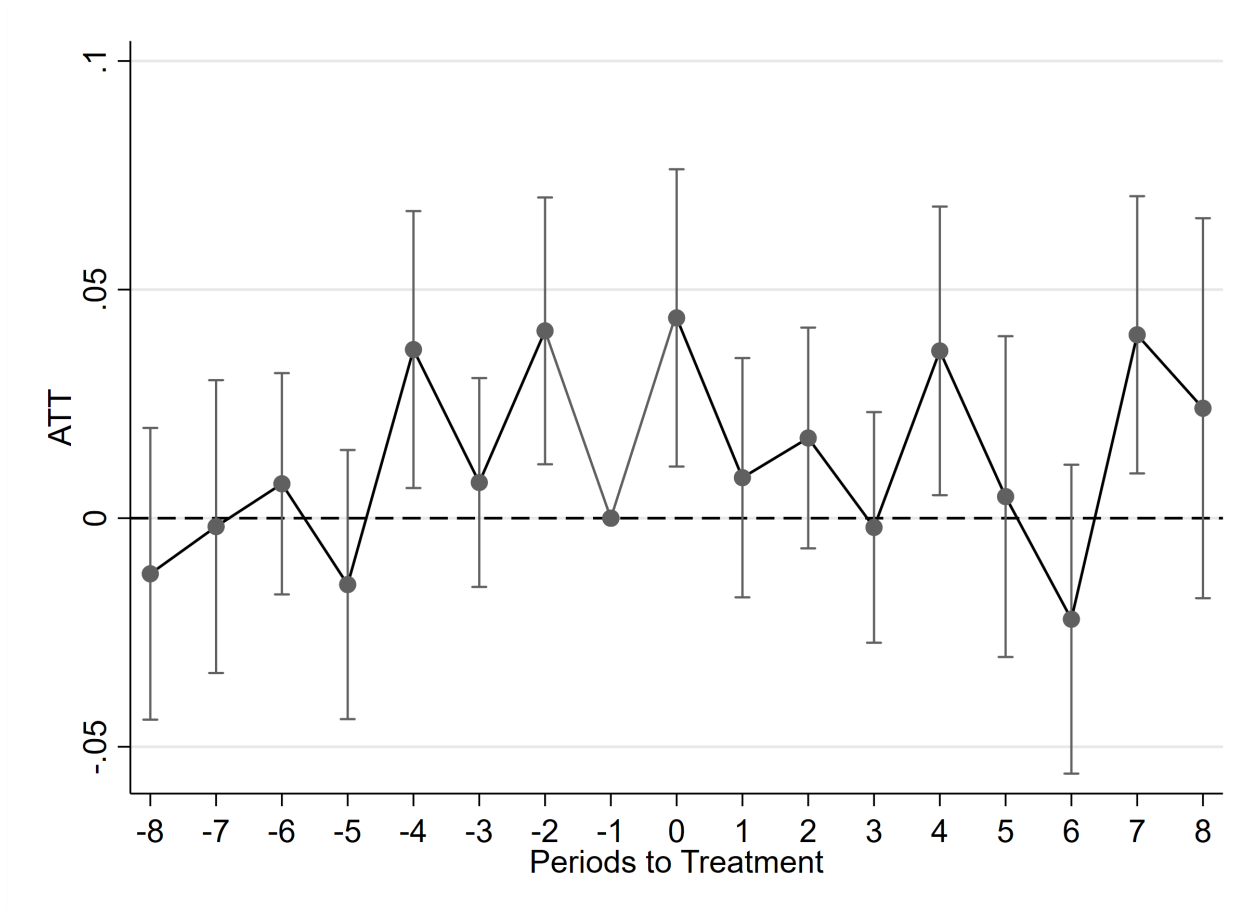


Figure C2: Event Study Estimates of the Effect of 3G Coverage on Internet Access, *csdid* Estimator

Notes: The figure reports event-study estimates of the effect of 3G coverage on an indicator for self-reported internet access at home, based on Equation 1 and estimated using the doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021) (*csdid*). The coefficients trace the dynamic effects relative to the period prior to the introduction of 3G coverage. Individual controls include demographic characteristics (age groups, college education, student status, gender, and marital status) and household composition variables, including the presence of young members (ages 15–34), college-educated or student household members, and counts of children (ages 0–17), working-age adults (ages 18–64), and elderly members (ages 65+).

All specifications are weighted using sampling weights, and standard errors clustered at the county level are reported in parentheses. The sample is restricted to urban individuals aged 18–64 and covers the period 2013Q2–2020Q3 (Iranian calendar years 1392–1399).

D Extended Literature Review

This section reviews the literature on the economic impacts of internet connectivity, with particular emphasis on the roll-out of mobile broadband (3G). I organize the discussion around four sets of outcomes: (i) aggregate and household economic outcomes, (ii) female labor force participation, (iii) labor-market matching, and (iv) political outcomes. The final strand is included because several papers use identification strategies that closely resemble the empirical approach in this paper.

A broad overview of this literature is provided by Hjort and Tian (2021), who survey evidence on how internet connectivity affects economic development in low- and middle-income countries. They emphasize that estimated effects vary across contexts, outcomes, and the type of connectivity studied, but that many studies find economically meaningful impacts, while other settings show limited or no effects.

D.1 Overall Economic Effects of the Internet

A central group of studies exploits spatial and temporal variation in mobile broadband roll-out linked to household panel surveys. Using operator coverage data and longitudinal household data, Bahia et al. (2021) and Bahia et al. (2020) estimate the welfare and labor-market impacts of mobile broadband expansion in Tanzania and Nigeria. Both papers document substantial gains in household consumption and significant reductions in poverty, and they argue that improvements in labor-market outcomes are an important mechanism behind these welfare effects.

Related evidence from Senegal is provided by Masaki et al. (2020), who combine household expenditure data with information on broadband infrastructure (including 3G coverage maps and the location of fiber-optic transmission nodes). They find positive effects of broadband access on consumption and poverty in both urban and rural areas, with larger impacts in urban settings, and provide suggestive evidence that labor-market improvements contribute to these gains. They also report limited evidence that fixed broadband has an independent effect on welfare outcomes.

Evidence from Brazil points to more limited aggregate employment effects but important heterogeneity. Using Brazil’s staggered internet rollout and matched employer–employee data, Dutz et al. (2017) find no statistically significant effect on aggregate employment, but document stronger employment growth in low-skilled, female-intensive occupations. In contrast, Hjort and Poulsen (2019) exploit the arrival of submarine internet cables and terrestrial backbone networks across Africa and find large positive employment effects. The gains are concentrated in higher-skill occupations, with smaller effects for less-educated workers.

D.2 Female Labor Force Participation

A number of studies examine whether internet access raises female labor force participation (LFP), generally finding positive effects. Using variation from the liberalization of broadband internet across provinces in Viet Nam, Chun and Tang (2018) estimate the causal impact of ICT adoption on firms' labor demand and find increased employment of female and college-educated workers, consistent with ICT reducing gender inequality in labor demand.

In the context of mobile broadband, Bahia et al. (2021) find that 3G coverage increases labor force participation and employment, with particularly strong gains among women. In Nigeria, Bahia et al. (2020) find more modest benefits for women on average, though they document occupational shifts among educated women exposed to mobile broadband and report that labor-market gains are concentrated among younger and educated men.

Evidence from Jordan highlights internet adoption as a channel affecting women's labor supply. Exploiting the rapid expansion of mobile broadband between 2010 and 2016, Viollaz and Winkler (2022) find that internet adoption increases female labor force participation. They further show that online job search accounts for part of the increase, while changes in traditional social norms among married women appear insufficient to explain the full effect.

D.3 Firm–Worker Matching

A second major mechanism emphasized in the literature is improved matching in the labor market. Using plausibly exogenous broadband expansion in Norway and rich administrative data on vacancies and job seekers, Bhuller et al. (2021) find that broadband internet increases both the speed and quality of matching, raising re-employment rates and starting wages and extending employment durations, consistent with a dominant matching channel.

Complementary evidence from the United States is provided by Kuhn and Mansour (2014), who show that unemployed individuals who search online are re-employed substantially faster than comparable workers who do not use online search. At the macro level, Lederman and Zouaidi (2020) document a negative association between indicators of digital-economy penetration and national unemployment rates across countries, though their evidence is correlational rather than causal.

These mechanisms may be particularly relevant in developing-country labor markets characterized by large search frictions. As noted by Hjort and Tian (2021), an important open question is how internet connectivity affects search and matching in environments where frictions are substantial and labor-market institutions differ from high-income settings.

D.4 Political Outcomes

A related literature studies the political consequences of mobile connectivity using identification strategies similar to those employed here. Leveraging global variation in the expansion of 3G coverage between 2008 and 2017, Guriev et al. (2021) estimate the effect of mobile broadband on confidence in government. In a related vein, Manacorda and Tesei (2020) exploit variation in mobile phone signal coverage to study political mobilization and protest participation.

Within-country studies also exploit subnational roll-out patterns of mobile broadband. Donati (2021) uses variation in the arrival of 3G in South Africa to estimate impacts on turnout, electoral competition, and protests, while Melnikov (2021) studies political polarization in the United States following the diffusion of 3G. Finally, Zhuravskaya et al. (2020) provides a comprehensive overview of this broader literature on the political effects of the internet and social media.

D.5 literature review

This paper contributes to the growing literature on the labor-market effects of internet access (Hjort and Tian, 2021; Hjort and Poulsen, 2019; Bhuller et al., 2023; Viollaz and Winkler, 2022; Dutz et al., 2017) as well as recent studies on mobile broadband expansion (Bahia et al., 2020, 2021; Masaki et al., 2020). The findings speak directly to a strand of this literature documenting larger employment gains for women following improvements in digital access (Dutz et al., 2017; Bahia et al., 2021; Viollaz and Winkler, 2022; Hjort and Tian, 2021). A more detailed discussion of the related literature is provided in Appendix D.

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